For this milestone, I selected my CS 300 Project Two course planner as my algorithms and data structures artifact. I originally created the project in October 2025. The program is written in C++ and uses a hash table to store course information such as the course title, number and prerequisites. It allows the user to load course data, print a list of courses in alphanumeric order, and search for a specific course and its prerequisites.

I chose this artifact because it shows my understanding of data structures and how they can be used to organize information. The original project already used a hash table with separate chaining, but I wanted to improve how the program handled the prerequisites. Originally, the program only displayed the direct prerequisite for a course. I enhanced it so the program can now follow the entire prerequisite chain.

I also improved the validation of the program. The enhanced version checks for missing course numbers and titles and makes sure prerequisites actually exist in the loaded course data. I also added a check for circular prerequisite relationships so the search doesn't continue forever if incorrect data creates a loop.

I believe I met the course outcome I planned to address this enhancement. My main goal was to demonstrate my ability to improve a computing solution using algorithms and data structures. The hash table continues to provide an organized way to store and search for courses, while the new prerequisite search adds more logic to the program. I do not have any major changes to my original outcome coverage plan because the enhancements I completed matched what I planned.

The biggest thing I learned from this enhancement was how I could improve an existing data structure without having to completely rebuild the program. One challenge was getting the program to continue through multiple levels of prerequisites instead of stopping the first one.

Using recursion helped solve this problem, but it also made me think about what would happen if the data was incorrect. Adding validations helped address those potential issues. Overall, this enhancement improved the original project and gave me more experience with hash tables, validation, and testing.